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BSSID based Monitoring Class Attendance System Using WiFi

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***Abstract*—Recent advancements in wireless technologies have evolved the growth of smart systems in day to day life. Nowadays, people are using WiFi connectivity for accessing the Internet or local hub inside home or office. A smart phone based checking system for monitoring class attendance through WiFi signal is developed in this work. Student, Teacher & Admin need to be installing the application. Most challenging part in this system is to make software design and send notification in smartphone. When Admin set specific time and students Smartphone connected with specific WiFi that time then they get notification for class attendance.**

Keywords—class attendance; fRF; Geeofencing;

I. INTRODUCTION

In every educational institution, checking student's attendance is very essential and compulsory rule to maintain that institution properly. Attendance checking is essential for knowing and ensuring every student's average attendance, study condition, other problems like illness or vacation etc. It shows a graphical representation of how many students are active in a class or in whole institution. In the past teachers or authorities used to call the name of students for the purpose of attendance. This is a time consuming process because the teacher has to call each person by name but when the number of student is huge then it is a great problem; just a waste of time and energy. It is a bad distraction for concentration. Nowadays smart phones are very available for both teachers and students. For this reason we are focusing on developing a system using mobile application. Both students and teachers will use this application for saving class time and their own purpose. When class time will be arrived, students will give their attendance by using this app. Student's attendance will store in database. Thus, student can get attendance mark automatically and teacher will monitor all this process and can easily know about total student's attendance and in how many class a student attended and how much mark he/she got, therefore, this working way of feature will bring more reliability to the proposed system.

II. LITERATURE SURVEY

Several types of smart attendance system were invented before. As example: Beacon based attendance system [1], Bluetooth based attendance [2], using QR for code scan for attendance [3][4][5]. However, this all system has some shortage for use. There are many smart systems which are developed based on expression recognition [6]. Brain Computer Interface (BCI) [7] framework needs EEG equipment. Need labor to proceed with the sight and sound. Hardware is expensive and tremendous. Our Software based framework does not require labor, does not bother with any sorts of equipment. Face recognition smart attendance system [8] cannot detect face if face distance is many more & when low light of image the system cannot work. In this Radio Frequency Identification based attendance system like as web based attendance, the framework advances a semi-mechanized methodology in catching the understudy participation, so there have many more cost [9-14]. In this finger print attendance system students press finger on this system when class is running so anybody can be fill disturb [15]. In our proposed system, user will not face these kinds of problems. The main advantage of this system is work on real time, if and only if the WiFi connection is appropriate.

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III. METHODOLOGY

A. Setup the Function

In this process, there are three types of work, here admin, teacher and students.

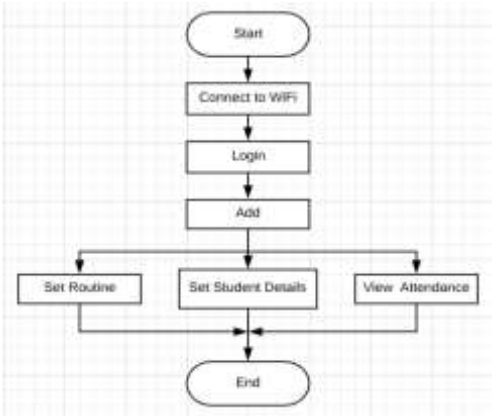

Fig. 1. Admin Workflow

The figure 1 shows that admin are working when admin connect a WiFi and then login the system. Admin can set up student details, delete this. They can also set the routine, which is followed by students and teachers, and admin can view attendance.

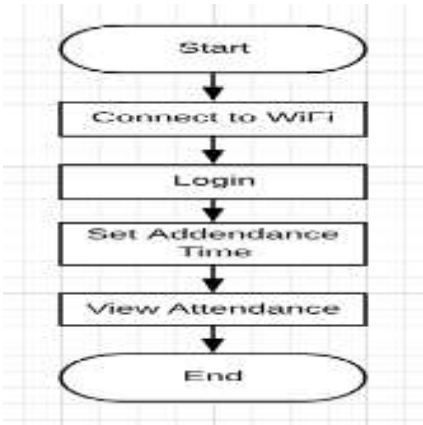

Fig. 2. Teachers Workflow

The figure 2 shows teacher firstly connect a specific WiFi and login the system then set a specific time. They have set starting and ending time to complete attendance. They can view the student attendance list. Students have to come in a specific classroom firstly in a particular subject. When teacher set the attendance time, students get a notification here showing that teacher's subject. Then they press the submit button and feedback a tools.

B. System workflow

Using smartphone device and router, all apps are working when they are connecting in WiFi. Only student app is working in specific router, which set by admin and it followed by routine. Teacher and Admin app are working any other network but student app does not work. Admin set security code and provide by students, when student install this app first need this security code. Otherwise, this app does not work. If admin can change the security code and if a student uninstalls this app after install this app one time and student put down the previous security code. This code does not work. This needed to be new security code. So reduce the proxy. In addition, attendance depend on teachers, only teachers know when set the time. Students are coming in the class and connecting the smartphone in specific WiFi. When teacher set attendance time all student get notification then just click on the submit button. Then get a tool your attendance is complete. Teachers see how many students are present in the class.

C. Requirements Analysis

We need mobile phone and Router to meet the hardware requirements as well as some software requirements like:

- Platform used for Android Application
- Platform: Android
- Language: Java, Xml
- Tools: Android Studio, Java SDK, Emulator
- Storage: Shared Preference, Firebase
- Design: Adobe XD
- Component: Activity, Fragment
- Android Studio

IV. ARCHITECTURE

In our proposed system there are three types of users, admin, teacher & students. Admin set the class time followed by routine, when students enter in the class then connect the WiFi. If "BSSID" match, then be notified which is shown in figure 3.

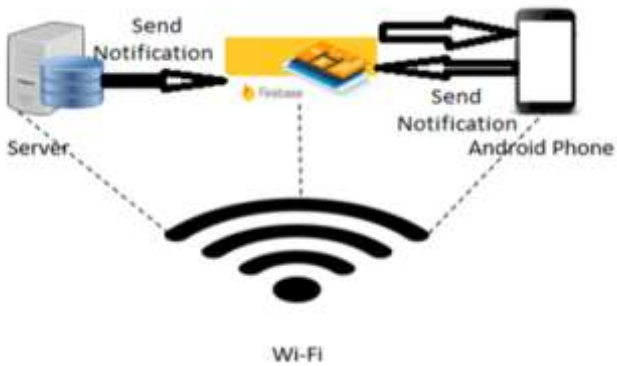

Fig. 3. Proposed model

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V. IMPLEMENTATION

In this portion, we will discuss the expected output of our proposed system. We have made three Apps to execute the system, which are connected with the same database and it controlled by admin. One App is for Admin, another is for teacher and other is for students.

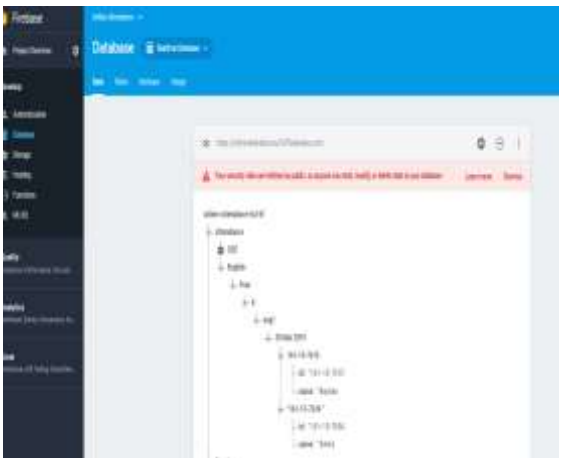

Fig. 4. Database of student's details

Admin have to sing up with email id and password to use the app shown in Fig. 5. Admin can add students with their Department, semester, name, id and section which is shown in Fig. 6, 7, 8 and can view students.

Fig. 5. Admin panel front page

Fig. 6. Admin setup routine detail in semester

Fig. 7. Admin setup student detail in semester

Fig. 8. Admin view student list

Admin also can set the class routine. To select routine, Admin have to set subject name, teacher's name, Department, Semester, Section, date shown in figure 9, 10. Lastly Admin have to set Security code for the students when they first time install their app and have to change the code for Security code for more secured is shown in figure 11.

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Fig. 9. Admin view the class routine

Fig. 10. Admin view routine a specific section

Fig. 11. Admin set security code

Second App is for the students. To open the app in first time they have to enter the valid Security code provided by Admin shown in figure 12. Students have to set their name, id, Department, Semester, and Section for one time in shown in figure 13. According their class routine if they stay in the classroom, their mobile connected with the same WiFi provided in the classroom, and teacher gives the starting and ending time for the attendance, when they open the app, they get notification with subject's name and teacher's name which is shown in figure 14.

Fig. 12. Registration for student

Fig. 13. Register for student and give attendance

They have to press on the button named "GIVE ATTENDANCE". After giving their attendance, they get a toast. Student can view their routine. To view routine, they have to select specific Day. After selecting all Day when they press the button, they can see the subject name, teacher name and class time shown in figure 15.

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Fig. 14. Student gets notification and gives attendance

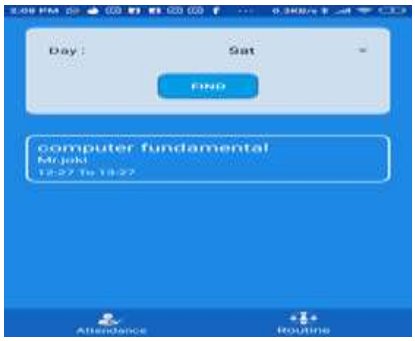

Fig. 15. Can view the class routine

This App is for the class teacher. To use this app, they have to sign up with their email id and password shown in figure 16. Firstly, they have to set the starting and ending time for the student when they get notification and give their attendance. By the app, they can see the total attended students in the class shown in figure 17. They have to select Department, Semester, Subject name, and current date. After selecting this entire information teacher can see the total scenario of the class, total present student and total absent student.

Fig. 16. Front page of teacher's app

Figure 17. Total present student in a class

To see the total attendance through semester of a student they have to select Department, Semester, subject name and id of the student.

VI. RESULT

In this system we find out the error and success accuracy. Here we are working in ten individual classes. Table 1 shows the accuracy result. Students' attendance checking system is shown in figure 18. We need powerful router and high configuration database to make the system error free.

Figure 18. Student attendance accuracy

Here, N= Number of students who present in class

A=how many students gets notification

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TABLE 1. Accuracy of our system

Total students	Present Student N	How many student get notification A	Error = ((N-A)/N)*100	Success Accuracy
45	33	29	12.12%	87.87%
40	38	35	7.89%	92.11%
40	29	25	13.79%	86.21%
35	25	23	8%	92%
40	34	31	8.89%	91.17%
44	32	29	9.37%	90.62%
40	35	33	5.72%	94.28%
40	36	32	11.11%	88.88%
40	23	23	0%	100%
23	23	20	13.04%	86.95%

VII. CONCLUSION AND FUTURE WORK

In this study, we have proposed and implemented a smart checking class attendance system using WiFi for educational institution. This system will save time, energy, stationery material and it will help to obey discipline. It will also provide an easy way to give attendance system by providing notification. Students will get notifications through an app and they will give their attendance in fixed classroom and time under a fixed network's area. Moreover, their attendance will store in database. Teachers will get student's attendance from database via teacher's app. Our class attendance system is very reliable for both teachers and students. After all, we may hope for a better solution of class attendance system. In future, we will use "Geofencing" instead of "BSSID" of router. So we can detect user inside or outside the classroom clearly.

We will add another option by which teachers will able to upload file, which will use by students. Moreover, students will upload file that can be seen by teacher. We will make student accuracy panel that only use admin panel and inform the parents via mobile app. Parents will get his/her children present accuracy in class. Hence, our system will have a highly reliable, secure and intelligent system to depend upon.

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